姓名:_____ 葉均承 應數一線性代數

學號: _____ **Quiz 8** 考試日期: 2026/11/4

不可使用手機、計算器，禁止作弊!

1. (10 points) Check if $T : \mathbb{R}^2 \to \mathbb{R}^2$ is an linear transformation, where $T([x, y]) = [x^2, y]$.

2. (15 points) Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation such that:

$$T([1, 0, 0]) = [2, -1], \quad T([0, 1, 0]) = [3, 4], \quad T([0, 0, 1]) = [0, 5]$$

   (a) Find the standard matrix A for this linear transformation T.
   Answer: $A =$ _____

   (b) Evaluate $T([2, -1, 3])$.
   Answer: $T([2, -1, 3]) =$ _____

3. (15 points) Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation that rotates vectors counterclockwise by an angle of $\frac{\pi}{3}$ $(60°)$.

   (a) Find the standard matrix representation A of the linear transformation T.
   Answer: $A =$ _____

   (b) Find the image of the vector $\vec{v} = [2, 2\sqrt{3}]$ under the transformation T.
   Answer: $T(\vec{v}) =$ _____